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Predicting Periodic and Chaotic Dynamics Using Neural Networks: A Study of the Logistic Map

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Abstract

Predicting the long-term evolution of chaotic dynamical systems remains a challenging problem due to their sensitivity to initial conditions. These study investigates how a neural networks accuracy in the multi-step prediction differs between periodic and chaotic regimes of logistic map. There are consider two different parameter $r = 3.2$ to the periodic and $r = 4.0$ the chaotic regime. Synthetic data was generated to train a multilayer perceptron (MLP) with two hidden layers of 32 neurons each. We evaluate it using the one-step and multi-step prediction accuracy. For multi-step prediction we use horizons of 1, 5, 10, 20 and 30 iterations. The baseline is the persistence model for one-step prediction. The performance is measured with mean absolute error (MAE) and root mean squared error (RMSE). Preliminary results show that the MLP method predicts one-step well. However, at a prediction of horizon of 30 steps, the MAE increases to 0.342372 in the chaotic regime, compared with 0.010647 in the periodic regime. This suggest that the accuracy of one-step does not imply a reliable long-term forecasting.

Keywords: logistic map; chaotic dynamics; periodic dynamics; neural networks; multi-step forecasting; machine learning.

0. Introduction

Dynamical systems provide mathematical framework for describe the evolution of a process over the time. They play a fundamental role in mathematics to model phenomena such as population growth, fluid motion, etc. However, even relatively simple deterministic models can exhibit complex dynamics, including periodic oscillations, bifurcations, and chaos [1].

A challenge of study a nonlinear dynamical systems is the prediction of chaotic behavior. Chaotic systems exhibit sensitive dependence on initial conditions. Thus, accurate long-term trajectory prediction may become difficult. This is related to amplification of small errors, which does not reliability of numerical and data-driven forecasting methods.

In recent years, machine learning techniques have been investigated as tools to approximate the evolution of nonlinear dynamical systems. Neural networks can learn a relationship of complex nonlinear systems from the dataset and generate predictions without solving the equation. Previous studies have explored approaches such as reservoir computing and recurrent neural networks for forecasting chaotic dynamics [2,3]. Although the challenge is maintaining the accuracy over longer prediction horizons remains challenging.

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The logistic map is a mathematical model for investigate nonlinear dynamical behavior. It is given by

$$x_{n+1} = rx_n(1 - x_n), \tag{1}$$

where $x_n \in [0, 1]$ represents the state of the system at iteration n, and $r \in [0, 4]$ is a control parameter.

The dynamic of the equation depend of the value r. The fixed points stability, periodic orbits, and chaotic dynamics depend of r [1]. Its mathematical simplicity with its rich dynamical behavior, makes that the logistic map interesting to study the relationship between stability and neural network prediction accuracy.

This investigation is about how multi-step prediction accuracy of a neural network differs between periodic and chaotic regimes of the logistic map. A multiplayer perceptron (MLP) is trained to approximate one-step and a multiple prediction horizons. The experiment consider $r = 3.2$ for periodic dynamic and $r = 4.0$ for chaotic dynamic. We hypothesize that the error in the chaotic regime as the forecasting horizon increases, the error becomes greater. This because the initial conditions sensitively. The main contribution of this work is give a reproducible experimental comparison of forecasting accuracy across those two dynamical regimes, with a mathematical explanation for this error propagation.

0.1. Research Question

The central research question of this study is:

How does multi-step prediction accuracy of a neural network differ between periodic and chaotic regimes of the logistic map?

0.2. Research Objective

The objective of this research is to evaluate the predictive performance of a multilayer perceptron neural network when approximating the evolution of the logistic map under periodic and chaotic regimes.

Particular attention is given to the effect of increasing the prediction horizon on forecasting accuracy.

0.3. Research Hypothesis

We hypothesize that a neural network may achieve low one-step prediction errors in both regimes, while recursive multi-step prediction errors will be substantially larger in the chaotic regime.

0.4. Research Contribution

The main contribution os this study is a reproducible experimental framework for evaluate how neural network can predict an dynamical system which have periodic regime and chaotic regime.

1. Materials and Methods

1.1. Mathematical Model: The Logistic Map

The logistic map is a discrete-time nonlinear dynamical system defined by

$$x_{n+1} = rx_n(1 - x_n), \tag{2}$$

where $x_n \in [0, 1]$ represents the state at iteration n, and $r \in [0, 4]$ is the control parameter.

Two parameter values are considered:

- $r = 3.2$: periodic regime.

- $r = 4.0$: chaotic regime.

Note that it is a deterministic maps, that is to say that given x_0 and r, every trajectory is determined by those. So, the map's behavior depends on r.

Let us study the dynamic of the map:

Note that a fixed point $x*$ is when

$$f_r(x^*) = x^*$$

Thus,

$$rx*(1-x*) = x* \quad \Rightarrow \quad x_1^* = 0 \quad \text{and} \quad x_2^* = 1 - 1/r \tag{3}$$

Since

$$f_r^{'} = r(1-2x)$$

and by fixed point theorem, we have that:

- $x_1^* = 0$ is a stable point when $0 \leq r < 1$
- $x_2^* = 1 - 1/r$ is a stable point when $1 < r < 3$

Therefore, when $r > 3$ the fixed point does not have stability and have a 2-periodic cycle.

Thus, $r = 3.2$ have a periodic regime.

Now, let us consider

$$x_{n+1} = 4x_n(1-x_n)\,, \quad r = 4 \tag{4}$$

and consider initial conditions x_0 and y_0 such that are arbitrary closer, it is to say,

$$|x_0 - y_0| < \varepsilon \qquad \text{for } \varepsilon > 0 \text{ small}$$

Then both trajectories begin closer but when we iterate the system, the trajectories can grow. Thus, consider x_n and y_n states of the system. By intermediate value theorem,

$$|f_r(x_n) - f_r(y_n)| = \left|f_r^{'}(e_n)\right||x_n - y_n|$$

for some e_n between x_n and y_n.

By Lyapunov interpretation when $r = 4$, we have that:

$$\lambda = \lim_{N\to\infty} \frac{1}{N} \sum_{n=0}^{N-1} ln\left|f_r^{'}(x_n)\right| = ln(2) > 0 \tag{5}$$

Furthermore, there is exponential sensitivity to small perturbations along the orbit. Therefore, when $r = 4$ is the chaotic regime.

1.2. Dataset Generation

The dataset was generated numerically from Equation (2).

For each parameter value, 100 trajectories were generated using randomly selected initial conditions from the interval $[0.05, 0.95]$.

Each trajectory contained 50 transitions, resulting in a total of 10,000 observations.

The input and target variables are defined as

$$\mathbf{X}_n = (x_n, r), \qquad y_n = x_{n+1}. \tag{6}$$

The dataset was divided into training and testing subsets using an 80/20 split at the trajectory level.

This procedure prevents observations from the same trajectory from appearing in both subsets.

To ensure reproducibility, a random seed of 42 was used for dataset generation and the training-test split. The dataset was divided as 80% for training set and 20% for test set. Before training, the inputs were standardized by 'StandardScaler' method. The scaler was fitted on the training data and applied to the test data to prevent data leakage.

Since the data was generated directed numerically from the deterministic logistic map, no missing-value or outlier removal was required.

1.3. Neural Network Architecture

A multilayer perceptron (MLP) was used to approximate the logistic map.

The neural network learns a function

$$f_\theta(x_n, r) \approx r x_n(1 - x_n), \tag{7}$$

where θ denotes the trainable parameters of the network.

The architecture consists of:

- Two input variables: x_n and r.
- Two hidden layers with 32 neurons each.
- Hyperbolic tangent activation.
- One scalar output predicting x_{n+1}.

The Adam optimizer was used for training, with a maximum of 1000 iterations and early stopping enabled. Input variables were standardized using a StandardScaler.

The neural network was implemented in Python using the `MLPRegressor` class.

The training objective was to minimize the mean squared error (MSE) between the predicted and the actual next states. The loss function is

$$\mathcal{L}(\theta) = \frac{1}{N} \sum_{i=1}^{N} (f_\theta(x_i, r_i) - y_i)^2.$$

Here, N denotes the number of training observations, y_i is the true next state, and $f_\theta(x_i, r_i)$ represents the prediction produced by the neural network.

The network parameters were optimized using the Adam algorithm, with a maximum of 1000 training iterations. By default, `MLPRegressor` reserves 10% of the training observations for internal validation. Training is stopped when the validation score fails to improve sufficiently for a specified number of consecutive iterations.

The independent test set was used exclusively for the final evaluation of the trained model.

1.4. Baseline Model

A persistence model was used as the baseline predictor:

$$\hat{x}_{n+1} = x_n. \tag{8}$$

This baseline assumes that the next state is equal to the current state.

Its performance was compared with the MLP using one-step predictions.

1.5. Multi-Step Prediction

Recursive forecasting was performed by repeatedly applying the trained neural network.

Starting from a known initial state $\hat{x}_0 = x_0$, predictions were generated by

$$\hat{x}_{n+1} = f_\theta(\hat{x}_n, r). \tag{9}$$

The prediction horizons considered were

$$h \in \{1, 5, 10, 20, 30\}. \tag{10}$$

For each regime, 30 initial conditions were used to evaluate the prediction error at different horizons.

1.6. Evaluation Metrics

Two error metrics were used.

The Mean Absolute Error (MAE) is

$$\mathrm{MAE} = \frac{1}{N} \sum_{i=1}^{N} |y_i - \hat{y}_i|. \tag{11}$$

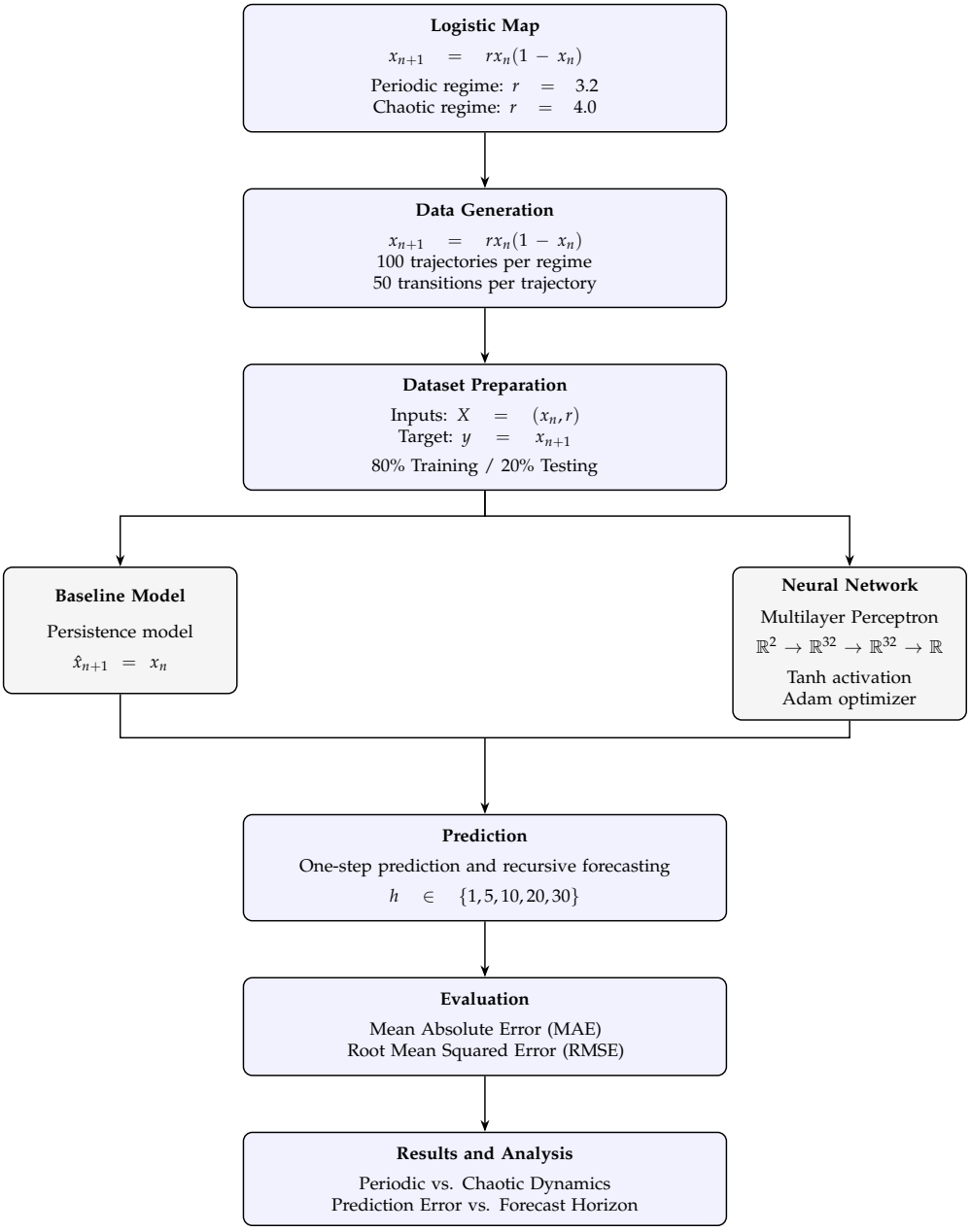
The Root Mean Squared Error (RMSE) is

$$\mathrm{RMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2}. \tag{12}$$

For a prediction horizon h, the average absolute error over M initial conditions is

$$E_h(r) = \frac{1}{M} \sum_{j=1}^{M} \left| x_h^{(j)} - \hat{x}_h^{(j)} \right|. \tag{13}$$

1.7. Experimental Workflow



height

Figure 1. Experimental workflow for predicting periodic and chaotic dynamics using a multilayer perceptron neural network.

2. Preliminary Results

2.1. One-Step Prediction Performance

The neural network was evaluated against the persistence baseline on the test set.

Table 1. One-step prediction performance.

Model	MAE	RMSE
Persistence Baseline	0.341258	0.398690
Neural Network MLP	0.003844	0.006417

The MLP achieved lower prediction errors than the persistence baseline, indicating that it successfully approximated the underlying nonlinear mapping in this experiment.

2.2. Multi-Step Prediction Performance

Table 2. MAE at different prediction horizons.

Horizon	Periodic ($r = 3.2$)	Chaotic ($r = 4.0$)
1	0.022974	0.004575
5	0.023725	0.091477
10	0.015485	0.439613
20	0.010633	0.412166
30	0.010647	0.342372

The results show that multi-step prediction errors remain relatively small in the periodic regime. In contrast, the chaotic regime exhibits substantially larger errors at longer prediction horizons.

When we are in the one step prediction, we have MAE of (0.022974) for periodic regime and (0.004575) for chaotic regimen that is lower than in the periodic regime. However, the behavior changes when we have more steps. Note that when we are in h=5, the MAE increases to 0.091477 for the chaotic regime compared with 0.023725 for the periodic regime.

The largest difference is when h=10, here the chaotic regimen reaches an MAE of 0.439613 compared with 0.015485 of the periodic regime. Also, the error at h=20 and h=30 in the chaotic regimen continues exhibit higher predictions. This behavior is consistent with the sensitivity to initial conditions of chaotic dynamical systems.

Although the neural network can approximate the one-step evolution of the logistic map, small approximation errors may be amplified through recursive predictions.

Also, note that the chaotic prediction error does not increase monotonically. After reaching 0.439613 at h = 10, the MAE decreases to 0.412166 at h = 20 and 0.342372 at h = 30. But, this does not contradict chaotic behavior.

Overall, these preliminary results suggest that accurate one-step predictions do not necessarily guarantee reliable of more steps predictions in chaotic dynamical systems.

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2.3. Graphical Analysis

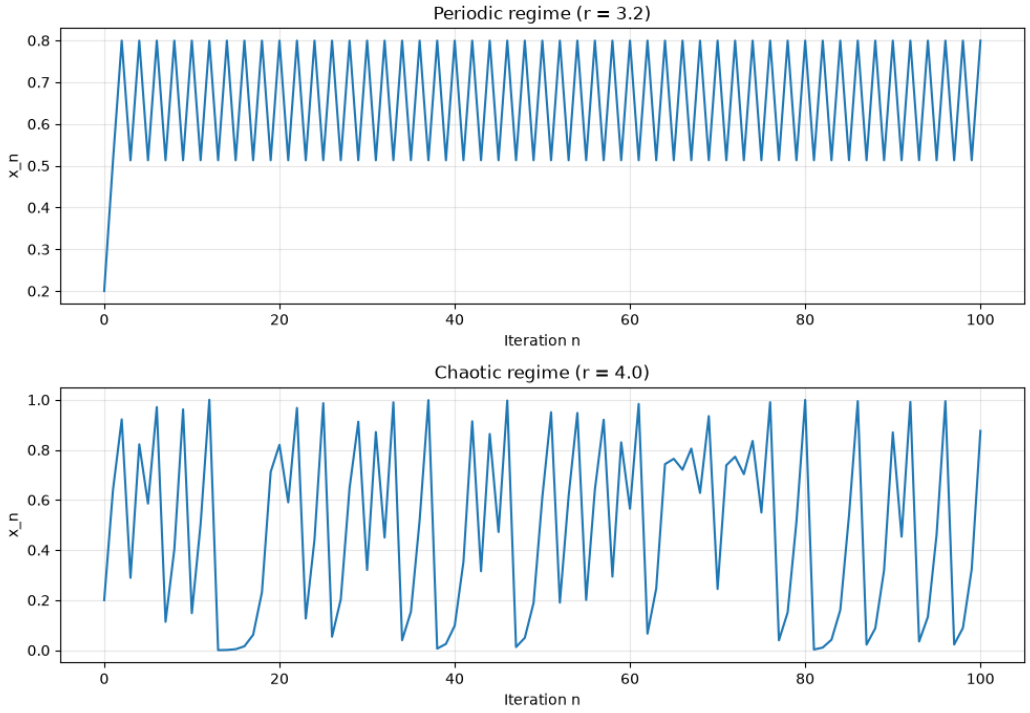


Figure 2. Periodic-Chaotic trajectories

Figure 2 illustrates the trajectories generated by the logistic map under periodic and chaotic regimes. For (r=3.2), the system approaches a stable period-two orbit. In contrast, for (r=4.0), the trajectory exhibits irregular fluctuations without an apparent repeating pattern, consistent with chaotic dynamics.

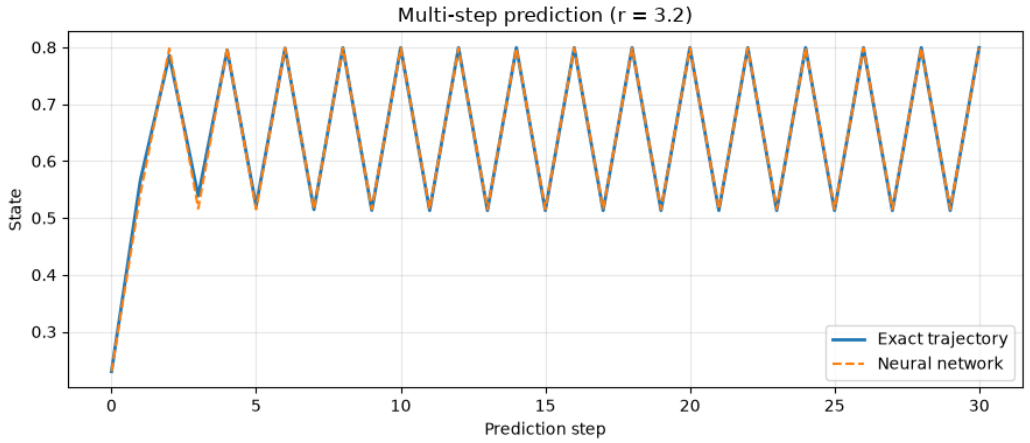


Figure 3. Multi-step prediction of periodic regime

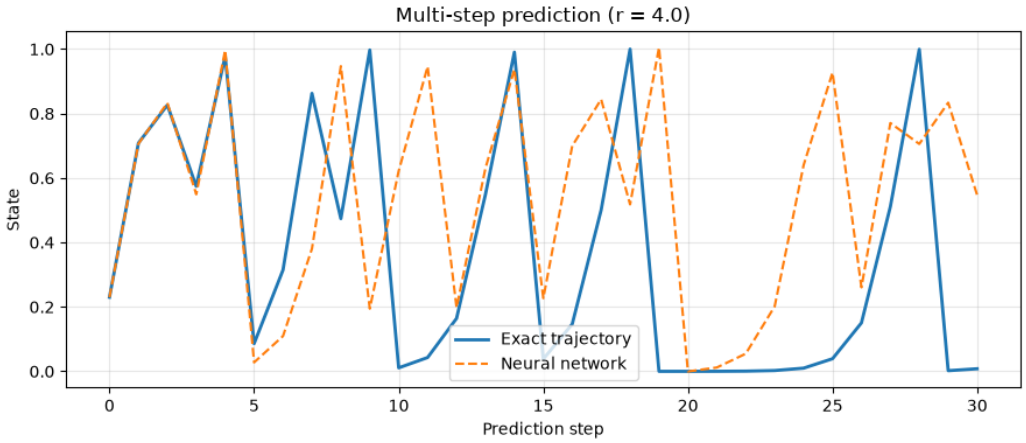


Figure 4. Multi-step prediction of chaotic regime

Figures 3 and 4 compare the exact trajectories with the recursive predictions produced by the neural network over 30 iterations. In the periodic regime, the predicted trajectory closely follows the exact solution. However, in the chaotic regime, the predictions initially approximate the true trajectory but progressively diverge as the prediction horizon increases. This illustrates the difficulty of maintaining accurate long-term trajectory predictions in chaotic systems.

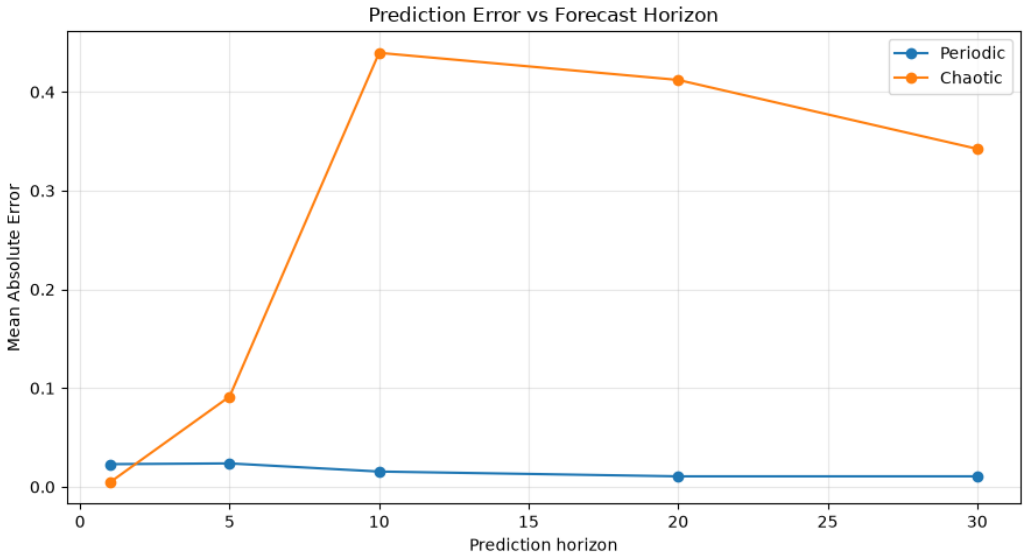


Figure 5. Errors

Finally, Figure 5 presents the mean absolute error as a function of the prediction horizon. The periodic regime have low prediction errors across all horizons. The chaotic regime have a increasing error reaching its greatest error at h=10. Also we can see that the chaotic predictions remains considerably higher that the periodic error.

3. Discussion

3.1. Interpretation of Preliminary Results

The preliminary results indicate that the multilayer perceptron (MLP) can approximates the one-step evolution of the logistic map. We have that the baseline have an MAE of 0.341258 and an RMSE of 0.398690, and the MLP improve this error which have an MAE of 0.003844 and an RMSE of 0.006417 on the test set. These suggest that the MLP learns the

nonlinear relationship between the current state, the control parameter, and the subsequent state.

However, that the predictions of one-step experiment have a low error, it does not imply that it will have a low error for a long-term forecasting. In fact, in the periodic regime with r=3.2 it have MAE of 0.010647 at a prediction horizon of 30 steps. In contrast, the chaotic regime when r=4 have an MAE of 0.342372 at the same horizon. This difference suggests that the dynamical properties of the system have influence in the reliability of recursive neural network predictions.

Note that, for the map

$$f_r(x) = rx(1-x),$$

the derivative is given by

$$f_r'(x) = r(1-2x).$$

For a small perturbation δ_n in the current state, a first-order approximation gives

$$\delta_{n+1} \approx f_r'(x_n)\delta_n.$$

Therefore, the evolution of small perturbations depends on the magnitude of the derivative along the trajectory.

In chaotic regimes, by the characterization of the Lyapunov exponent, the perturbations grows exponentially. Therefore, in the neural network the prediction error can be amplified during the recursive forecasting.

In periodic regimes, the logistic map have a period-two orbit that is a attraction for the system. This provides stability against small perturbations. That why the neural network have good predictions over longer horizons.

In contrast, chaotic regime exhibit dependence on initial conditions, that make that the long-term trajectory predictions is more difficult to predict. Nevertheless, since the distance between real and predicted trajectories can fluctuate as the system evolves, the error does not necessary increase monotonically.

3.2. Limitations

First, we only choose two values of r to do the experiment, with r=3.2 as a representative of the periodic-two regime and with r=4 for the chaotic regimen. Thus, the experiment can not provide a complete characterization of the transition from periodic to chaotic dynamics. Also, since we only do the experiment with those r chosen, this limits our conclusions about the study of general dynamical regimens.

Second, we only choose one architecture of MLP, with two hidden layers of 32 neurons each. Therefore, other architecture of neural network can improve or produce different results. Thus, we can not generalized of other machine learning models.

Third, the dataset was generated synthetically used the logistic map. Although this data set provides a environment that is controlled without measurement noise, but the real world dynamical system often contain noise.

Finally, the baseline was apply only for the one-step analysis. Then, we do not have a real comparison with more horizons. Thus, future experiments should include additional baseline models and evaluate their performance under the same conditions.

3.3. Future Work

Future work will focus on extending the experimental framework to gain a true understanding of how neural networks operate within a nonlinear dynamic system.

First, we will consider more logistic map parameters to investigate the transition from periodic to chaotic behavior. Thus, it will give us more information to evaluate how prediction accuracy changes across different dynamical regimes.

Second, We can implement different neural architectures to compare with MLP. Also, we can implement a better baseline to provide more comprehensive evaluation of forecasting performance.

Third, we will improve the data set with incorporate noisy observations and a larger number of initial conditions to investigate the robustness and generalization of the model. Also, we will use longer horizons to evaluate the limitations of recursive forecasting.

Finally, we will investigate the relation between dynamical instability and prediction error, with a mathematical analysis based on Lyapunov exponents.

This work will provide more detailed assessment of the relation between chaotic dynamics and machine learning prediction.

4. Conclusions

In this study we investigate the predictive performance of a multilayer perceptron neural network in periodic and chaotic regimes of the logistic map. This study consider two parameter values r=3.2 representing the periodic dynamics and r=4.0 representing the chaotic dynamics. We generated synthetically data set, and train the neural network was training with those and evaluated through one-step and recursive multi-step forecasting, using MAE and RMSE.

The preliminary results show that the neural network can approximate the one-step evolution of the logistic map achieving and MAE of 0.003844 and an RMSE of 0.006417. However, we see that considering longer prediction horizons, the prediction of chaotic regime is not well. These findings indicate that recursive predictions remain substantially more accurate in periodic regime than in the chaotic regime under the experimental conditions.

The results are consistent with the mathematical properties of the nonlinear dynamical systems. The study suggests that accurate one-step approximation does not guarantee long-term trajectory predictions. Nevertheless, this conclusions are limited to the selected parameter values, neural networks architecture, and experimental configuration.

Data Availability Statement: The Python code used for synthetic dataset generation, neural network training, baseline evaluation, and multi-step forecasting is publicly available in the GitHub repository: https://github.com/MilenaF199/logistic-map-neural-network.

The repository contains the source code, required Python dependencies, and graphical results of the experiments. The dataset can be regenerated using the provided code and the experimental configuration described in this manuscript.

Abbreviations

The following abbreviations are used in this manuscript:

MLP	Multilayer Perceptron
MAE	Mean Absolute Error
RMSE	Root Mean Squared Error
RNN	Recurrent Neural Network
LSTM	Long Short-Term Memory

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